

Example: Microsatellite data set

POPULATION SIZE, MIGRATION, DIVERGENCE, ASSIGNMENT, HISTORY

Bayesian inference using the structured coalescent

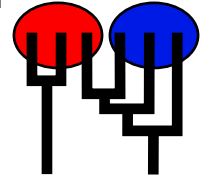
Migrate-n version 5.0.7 [May-01-2025]

Compiled for PARALLEL computer architectures

One master and 1 compute nodes are available.

Program started at Mon Jul 7 19:53:00 2025

Program finished at Mon Jul 7 19:53:20 2025 [Runtime:0000:00:00:20]



Options

Datatype: Microsatellite data [Brownian motion]
Missing data: not included

Inheritance multipliers in use for Thetas:
All loci use an inheritance multiplier of 1.0

Data set was subsampled: used a random sample of size:
5 and seed 13

Random number seed: (from parmfile) 310705631

Start parameters:

Theta values were generated Using a percent value of the prior

M values were generated Using a percent value of the prior

Connection matrix:

m = average (average over a group of Thetas or M,

s = symmetric migration M, S = symmetric 4Nm,

0 = zero, and not estimated,

* = migration free to vary, Thetas are on diagonal

d = row population split off column population, D = split and then migration

Population	1	2
1 population_num	*	0
2 population_num	D	*

1 population_numb growing/shrinking
 2 population_numb constant size

Order of parameters:

1	Θ_1	<displayed>
2	Θ_2	<displayed>
3	$M_{1 \rightarrow 2}$	<displayed>
4	$\Delta_{1 \rightarrow 2}$	<displayed>
5	$\sigma_{1 \rightarrow 2}$	<displayed>
6	g_1	<displayed>

Mutation rate among loci:

Mutation rate is constant for all loci

Analysis strategy:

Bayesian inference

-Population size estimation:

Exponential Distribution

-Population growth estimation: Growth [Exponential]

-Geneflow estimation:

Exponential Distribution

-Divergence time estimation:

Normal Distribution (mean and standard dev.)

Proposal distributions for parameter

Parameter	Proposal
Theta	Slice sampling
M	Slice sampling
Divergence	Metropolis sampling
Divergence Spread	Metropolis sampling
Growth	Metropolis sampling
Genealogy	Metropolis-Hastings

Prior distribution for parameter

Parameter	Prior	Minimum	Mean*	Maximum	Delta	Bins	UpdateFreq	
1 Theta	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
2 Theta	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
3 M	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
4 Splittime mean	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
5 Splittime std	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
6 Growth	* *	Uniform	-100.000000	0.000	100.0	20.000000	1500	0.08333

[* * means priors were set globally]

Posterior distribution:

Parameter values were collected using MCMC, these values were then used to generate the posterior histograms using KERNEL SMOOTHING (window=41) and subsequent SAVITZKY-GOLAY SMOOTHING (window=41) for combination over loci

Markov chain settings:

	Long chain
Number of chains	1
Recorded steps [a]	200
Increment (record every x step [b])	1
Number of concurrent chains (replicates) [c]	2
Visited (sampled) parameter values [a*b*c]	400
Number of discard trees per replicate (burn-in * b)	50

Multiple Markov chains:

Static heating scheme

1000000.00 4 chains with temperatures 3.00 1.50 1.00
Swapping interval is 1

Print options:

Data file: infile.msat
parmfile.msat
Haplotyping is turned on: NO
Output file: outfile_msat
Posterior distribution raw histogram file: bayesfile_msat
Raw data from the MCMC run: bayesallfile_msat.gz
Print data: No
Print genealogies [only some for some data type]: None

Data summary

Data file: infile.msat
 Datatype: Microsatellite data [Brownian]
 [Data was used as repeat-length information]
 Number of loci: 10

Mutationmodel:

Locus	Sublocus	Mutationmodel	Mutationmodel parameters
1	1	Brownian Motion	[none]
2	1	Brownian Motion	[none]
3	1	Brownian Motion	[none]
4	1	Brownian Motion	[none]
5	1	Brownian Motion	[none]
6	1	Brownian Motion	[none]
7	1	Brownian Motion	[none]
8	1	Brownian Motion	[none]
9	1	Brownian Motion	[none]
10	1	Brownian Motion	[none]

Data set was subsampled: used a random sample of size: 5

Population	Locus	Gene copies data	(missing)
1 population_number__0	1	50	(0)
	2	50	(0)
	3	50	(0)
	4	50	(0)
	5	50	(0)
	6	50	(0)
	7	50	(0)
	8	50	(0)
	9	50	(0)
	10	50	(0)
2 population_number__1	1	42	(0)
	2	42	(0)
	3	42	(0)
	4	42	(0)
	5	42	(0)
	6	42	(0)
	7	42	(0)

Total of all populations	8	42	(0)
	9	42	(0)
	10	42	(0)
	1	92	(0)
	2	92	(0)
	3	92	(0)
	4	92	(0)
	5	92	(0)
	6	92	(0)
	7	92	(0)
	8	92	(0)
	9	92	(0)
	10	92	(0)

Bayesian Analysis: Posterior distribution table

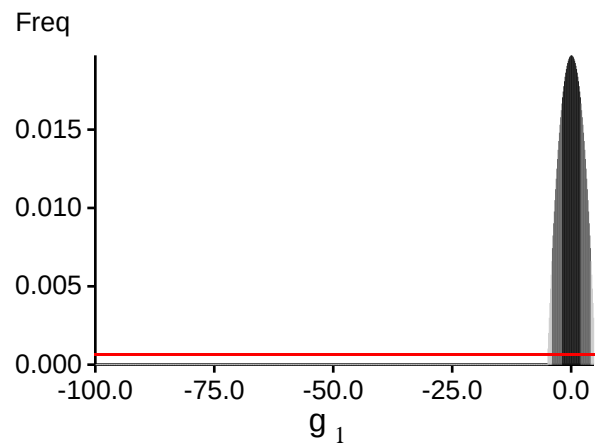
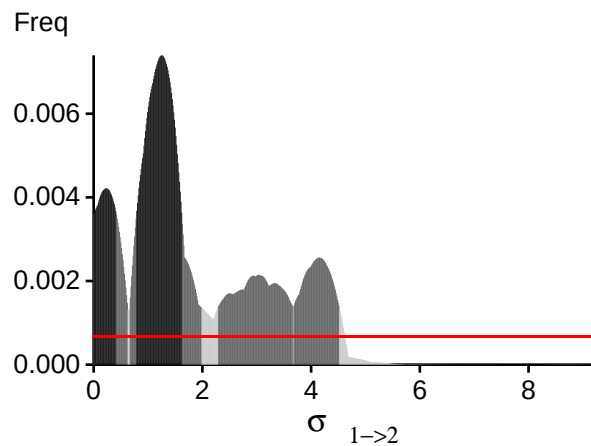
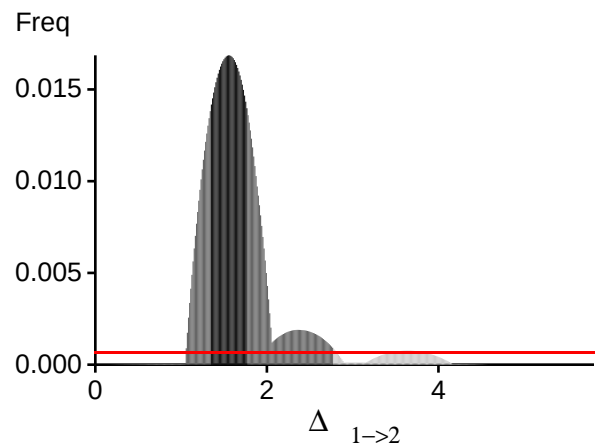
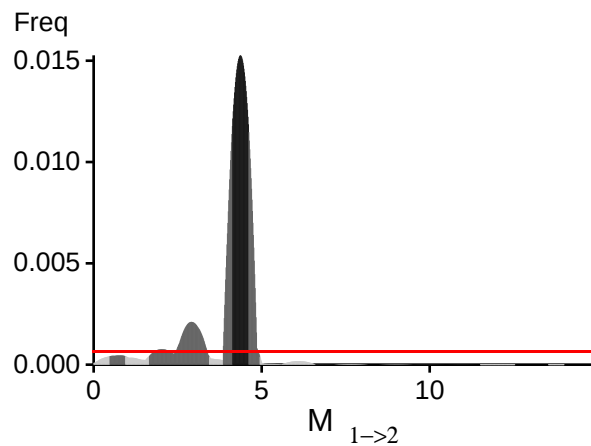
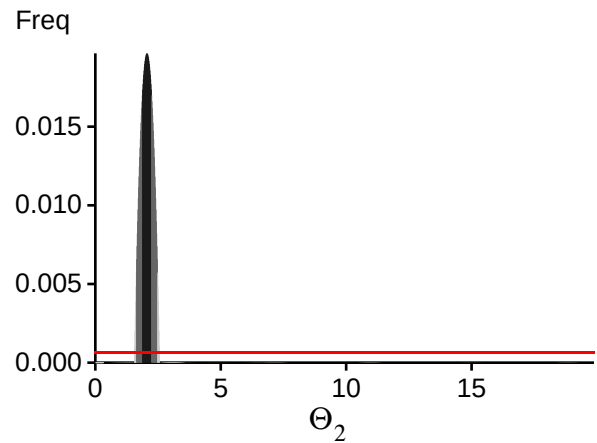
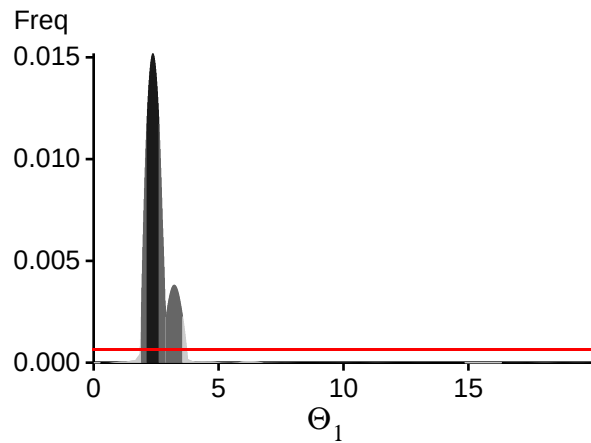
Locus	Parameter	2.5%	25.0%	Mode	75.0%	97.5%	Median	Mean
1	Θ_1	1.72000	2.05333	2.66000	3.26667	4.48000	3.71333	8.70993
1	Θ_2	2.64000	2.89333	3.39333	3.89333	4.10667	6.11333	5.79528
1	$M_{1 \rightarrow 2}$	1.013	1.453	2.220	2.600	3.360	2.100	2.270
1	$D_{1 \rightarrow 2}$	1.50667	1.94667	2.39333	2.86667	4.44000	2.60667	2.77205
1	$S_{1 \rightarrow 2}$	0.00000	0.00000	0.15333	0.61333	0.61333	0.82000	0.73900
1	g_1	-4.13333	-1.86667	0.20000	2.13333	4.40000	0.33333	0.19291
2	Θ_1	0.68000	0.93333	1.27333	1.52000	3.50667	2.10000	2.25472
2	Θ_2	8.82667	9.22667	9.94000	10.56000	11.18667	9.88667	10.04600
2	$M_{1 \rightarrow 2}$	2.280	2.733	3.073	3.427	5.133	3.540	3.558
2	$D_{1 \rightarrow 2}$	0.00000	0.10667	0.42000	0.77333	1.40000	2.43333	1.99724
2	$S_{1 \rightarrow 2}$	0.00000	0.06667	0.39333	0.96000	1.38667	3.24667	2.54587
2	g_1	-4.93333	-2.26667	-0.20000	1.60000	4.13333	-0.06667	-0.26932
3	Θ_1	4.14667	5.58667	6.14000	6.60000	7.00000	6.15333	8.15331
3	Θ_2	1.12000	1.24000	1.83333	2.88000	3.16000	6.84667	9.21171
3	$M_{1 \rightarrow 2}$	0.173	4.027	4.420	4.853	7.067	3.220	3.400
3	$D_{1 \rightarrow 2}$	0.00000	0.25333	0.66000	1.05333	2.89333	1.88667	2.28572
3	$S_{1 \rightarrow 2}$	0.33333	2.66667	3.08667	4.29333	5.04000	3.31333	3.51879
3	g_1	-3.73333	-1.20000	0.86667	2.66667	5.33333	1.00000	0.81812
4	Θ_1	2.94667	3.05333	3.59333	4.13333	4.28000	6.19333	7.01494
4	Θ_2	2.66667	4.06667	4.64667	4.97333	5.30667	9.44667	8.82798
4	$M_{1 \rightarrow 2}$	0.000	0.000	0.313	0.653	0.947	2.167	1.876
4	$D_{1 \rightarrow 2}$	0.70667	1.28000	1.91333	2.36000	4.06667	2.04667	2.21037
4	$S_{1 \rightarrow 2}$	0.72000	2.52000	2.75333	3.01333	3.36000	2.16667	2.20082
4	g_1	-6.40000	-2.13333	-0.33333	2.26667	5.46667	0.06667	-0.30039
5	Θ_1	0.88000	1.10667	1.48667	1.88000	3.89333	2.42000	2.61529
5	Θ_2	0.54667	0.70667	1.14000	2.36000	2.81333	2.28667	4.65178
5	$M_{1 \rightarrow 2}$	0.840	4.480	4.900	5.320	6.387	3.593	3.928
5	$D_{1 \rightarrow 2}$	1.06667	1.44000	1.83333	2.32000	3.72000	2.08667	2.21899
5	$S_{1 \rightarrow 2}$	0.00000	0.42667	1.02000	1.58667	1.97333	2.76667	2.73018
5	g_1	-4.26667	-1.86667	0.06667	1.86667	4.26667	0.20000	0.00465
6	Θ_1	1.89333	3.46667	3.91333	4.26667	4.56000	3.54000	3.43140

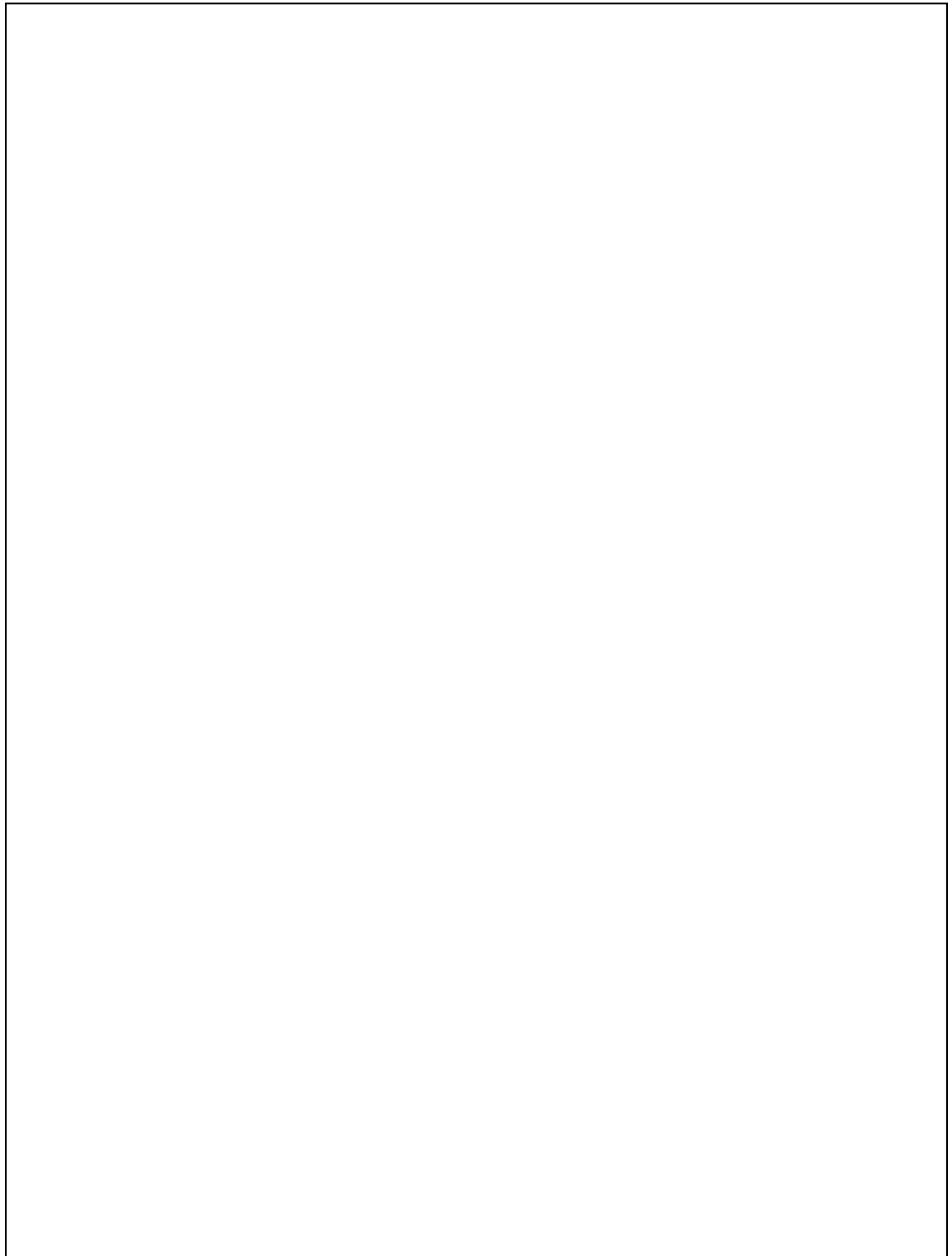
Locus	Parameter	2.5%	25.0%	Mode	75.0%	97.5%	Median	Mean
6	Θ_2	0.58667	1.38667	2.02000	2.41333	3.52000	4.27333	6.32133
6	$M_{1 \rightarrow 2}$	0.813	3.840	4.287	4.747	4.933	3.607	3.839
6	$D_{1 \rightarrow 2}$	0.20000	0.52000	0.76667	1.00000	1.38667	1.82000	1.66499
6	$S_{1 \rightarrow 2}$	0.00000	0.53333	0.90000	1.25333	1.54667	2.78000	2.37015
6	g_1	-4.40000	-2.00000	-0.06667	1.73333	4.13333	0.06667	-0.13442
7	Θ_1	0.64000	0.72000	2.62000	3.28000	3.36000	2.30000	2.33993
7	Θ_2	2.16000	2.57333	3.23333	3.92000	7.21333	5.55333	8.28692
7	$M_{1 \rightarrow 2}$	0.000	0.427	0.687	0.933	1.440	0.727	0.786
7	$D_{1 \rightarrow 2}$	1.81333	2.29333	2.66000	3.26667	5.86667	3.38000	3.56333
7	$S_{1 \rightarrow 2}$	0.00000	0.06667	0.32667	0.53333	1.38667	1.27333	1.24264
7	g_1	-4.26667	-1.86667	0.06667	1.86667	4.26667	0.20000	0.00000
8	Θ_1	2.12000	2.53333	2.90000	3.33333	7.41333	4.62000	5.87809
8	Θ_2	1.16000	3.29333	3.68667	3.98667	4.29333	4.70000	6.71150
8	$M_{1 \rightarrow 2}$	4.947	5.653	6.100	6.493	7.480	4.020	4.119
8	$D_{1 \rightarrow 2}$	1.05333	1.28000	1.66000	2.04000	2.50667	2.20667	2.53348
8	$S_{1 \rightarrow 2}$	1.09333	1.25333	1.67333	2.21333	2.82667	1.91333	2.02820
8	g_1	-3.86667	-1.20000	0.73333	2.53333	5.20000	0.86667	0.73881
9	Θ_1	1.48000	2.56000	3.11333	3.48000	4.24000	2.99333	2.96940
9	Θ_2	7.34667	7.57333	8.08667	8.61333	8.77333	10.80667	11.16532
9	$M_{1 \rightarrow 2}$	0.160	0.800	1.153	1.347	3.920	2.513	4.542
9	$D_{1 \rightarrow 2}$	0.64000	1.42667	1.72667	2.02667	3.61333	2.43333	2.59873
9	$S_{1 \rightarrow 2}$	0.65333	1.50667	1.91333	2.57333	3.74667	2.12667	2.23039
9	g_1	-4.26667	-1.86667	0.06667	1.86667	4.13333	0.20000	-0.00849
10	Θ_1	2.09333	3.22667	3.67333	4.14667	5.45333	4.16667	5.91859
10	Θ_2	0.94667	1.16000	1.78000	2.72000	5.28000	2.70000	4.33453
10	$M_{1 \rightarrow 2}$	0.360	0.693	0.953	1.200	3.173	1.993	1.998
10	$D_{1 \rightarrow 2}$	0.00000	0.12000	0.84667	1.20000	1.42667	1.10000	1.64993
10	$S_{1 \rightarrow 2}$	1.86667	2.88000	3.35333	3.77333	4.80000	3.23333	2.98941
10	g_1	-4.80000	-1.86667	0.20000	2.00000	4.93333	0.33333	0.14992
All	Θ_1	1.89333	2.12000	2.36667	2.61333	2.89333	2.47333	2.58416
All	Θ_2	1.62667	1.86667	2.06000	2.24000	2.48000	2.07333	2.07770
All	$M_{1 \rightarrow 2}$	3.840	4.120	4.367	4.613	4.947	4.300	3.982
All	$D_{1 \rightarrow 2}$	1.04000	1.33333	1.55333	1.77333	2.77333	1.62000	1.72638
All	$S_{1 \rightarrow 2}$	0.65333	0.77333	1.26000	1.62667	1.98667	1.44667	1.77421
All	g_1	-4.13333	-2.00000	0.06667	2.00000	4.13333	0.20000	0.06667

Citation suggestions:

- Beerli P., 2006. Comparison of Bayesian and maximum-likelihood inference of population genetic parameters. *Bioinformatics* 22:341-345
- Beerli, P., H. Ashki, S. Mashayekhi, and M. Palczewski, 2022. Population divergence time estimation using individual lineage label switching. *G3 Genes & Genomes & Genetics*, 12(4), 02 2022.
- Beerli P., 2009. How to use MIGRATE or why are Markov chain Monte Carlo programs difficult to use? In *Population Genetics for Animal Conservation*, G. Bertorelle, M. W. Bruford, H. C. Hauffe, A. Rizzoli, and C. Vernesi, eds., vol. 17 of *Conservation Biology*, Cambridge University Press, Cambridge UK, pp. 42-79.
- Beerli, P., S. Mashayekhi, M. Sadeghi, M. Khodaei, and K. Shaw, 2019. Population genetic inference with migrate. *Current Protocols in Bioinformatics*, 68(1):e87.

Bayesian Analysis: Posterior distribution over all loci





Log-Probability of the data given the model (marginal likelihood)

Use this value for Bayes factor calculations:

$BF = \text{Exp}[\ln(\text{Prob}(D \mid \text{thisModel})) - \ln(\text{Prob}(D \mid \text{otherModel}))]$

or as $LBF = 2 (\ln(\text{Prob}(D \mid \text{thisModel})) - \ln(\text{Prob}(D \mid \text{otherModel})))$

shows the support for thisModel]

Locus	TI(1a)	BTI(1b)	HS(3)
1	-3997.31	-768.67	-359.18
2	-5087.58	-853.68	-77.90
3	-16604.33	-2699.48	-85.61
4	-580.86	-328.72	-1874.36
5	-868.20	-164.70	-38.14
6	-11154.05	-1894.84	-260.98
7	-251.70	-86.36	-137.20
8	-7576.71	-1274.81	-169.26
9	-463.32	-131.60	-231.46
10	-276.00	-97.34	-113.95
All	-46791.06	-8231.20	-3279.02

(1a) TI: Thermodynamic integration: $\log(\text{Prob}(D \mid \text{Model}))$: Good approximation with many temperatures

(1b) BTI: Bezier-approximated Thermodynamic integration: when using few temperatures USE THIS!

(3) HS: Harmonic mean approximation: Overestimates the marginal likelihood, poor variance

[Scaling factor = 69.014149]

Citation suggestions:

Beerli P. and M. Palczewski, 2010. Unified framework to evaluate panmixia and migration direction among multiple sampling locations, *Genetics*, 185: 313-326.

Palczewski M. and P. Beerli, 2014. Population model comparison using multi-locus datasets.

In M.-H. Chen, L. Kuo, and P. O. Lewis, editors, *Bayesian Phylogenetics: Methods, Algorithms, and Applications*, pages 187-200. CRC Press, 2014.

Acceptance ratios for all parameters and the genealogies

Parameter	Accepted changes	Ratio
Θ_1	317/317	1.00000
Θ_2	358/358	1.00000
$M_{1 \rightarrow 2}$	330/330	1.00000
$\Delta_{1 \rightarrow 2}$	321/349	0.91977
$\sigma_{1 \rightarrow 2}$	313/340	0.92059
g_1	28/344	0.08140
Genealogies	949/1962	0.48369

MCMC-Autocorrelation and Effective MCMC Sample Size

Parameter	Autocorrelation	Effective Sample Size
Θ_1	0.94404	1153.72
Θ_2	0.92709	1515.73
$M_{1 \rightarrow 2}$	0.92494	1560.36
$\Delta_{1 \rightarrow 2}$	0.95455	930.72
$\sigma_{1 \rightarrow 2}$	0.94772	1080.34
g_1	-0.00003	39802.02
Genealogies	0.92834	1524.61

Average temperatures during the run

Chain	Temperatures
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1	1.00000
2	1.00000
3	1.00000
4	1.00000

Potential Problems

This section reports potential problems with your run, but such reporting is often not very accurate. With many parameters in a multilocus analysis, it is common that some parameters for some loci will not be informative. These parameters then trigger suggestions to increase the prior range that are not sensible. Do not blindly follow the suggestions given. If some parameters are flagged, inspect the tables carefully and judge whether an action is required. Suppose you run a Bayesian inference with sequence data for macroscopic species. In that case, there is rarely the need to increase the prior for Theta beyond 0.1. If you use microsatellites data, it is rather common that your prior distribution for Theta should have a range from 0.0 to 100 or more. With many populations (>3), it is also very common that some migration routes are estimated poorly because the data contains little or no information for that route. Increasing the prior range will not help in such situations, but reducing the number of parameters may help.

No warning was recorded during the run